

STAT8020: Project for the academic year 2025-2026

Students are required to do a group project in this course. Each group will comprise 2 to 4 students. Students should inform Dr. Eric Li on group membership on or before the test date. Solo projects are also acceptable but you need to inform Eric.

Suggested project topic:

Each group will devise one or several intraday trading algorithms for Hang Seng Index futures contract. You can either use your own favorable intraday trading methods, methods published in newspapers, magazines, books or academic journals, or you can use black-box methods that may involve modern AI techniques.

Minute-by-minute data will be provided to you in moodle a little bit later (filename: ***hi1_20170701_20200609.csv***) and you are required to back-test your algorithms with this historical dataset. In this dataset, for eg, 20170701 means July 1 2017, time 235700 means 23:57:00 in the evening. Note that hi1 means 1-month HSI futures contract and its trading hrs are much longer than the stock market. You can devise your own in-sample and out-sample periods to perform back-testing and report on back-testing results in a project report. In the report, you are requested to recommend one algo (or several algos) for trading in future period. The recommended algo or algos need to have well specified parameters which can be implemented in real trading.

Alternative project topics:

The topic above is only a suggested project. If you are interested in algo's other than intraday trading algo's using data/markets other than hi1, pls feel free to contact Eric for your project feasibility.

The deadline for uploading the report to Moodle is ***11:59pm Sunday May 31, 2026***.

Your report should include the following:

1. Introduction
2. Detailed description of your trading algo and its rationale
3. Specification of in-sample, out-sample periods
4. Back-testing results and performance characteristics
5. Optimization of trading parameters if there is any
6. Comments on slippages which may occur in real trading
7. Implementation difficulty and problems that may arise from real time trading
8. Risk management features for implementation
9. Whatever you think is relevant
10. References
11. Conclusion

Whether your recommended algo (or algos) is profitable or not in year 2026 is just for your own curiosity and it will bear no relevance to the grading of the project. The evaluation criteria will

be on presentation, innovation, sound reasoning, comprehensiveness, understanding of the market and clever observations. Also, if there is any group member (or members) whose contribution to the project is particularly valuable, please mention it in the report.